THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET A

Chapters 5 & 7: **Arithmetic Progressions, Coordinate Geometry** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The common difference of the AP $0.6, 1.7, 2.8, \ldots$ is: 1

(a) 1.7 **(b)** −1.1 **(c)** 1.1 **(d)** 0.6

2. The 20th term of the AP $4, 9, 14, \ldots$ is: 1

(a) 99 **(b)** 100 **(c)** 104 **(d)** 94

3. Which term of the AP $2, 6, 10, \ldots$ is 50? 1

(a) 25th **(b)** 12th **(c)** 13th **(d)** 14th

4. A student states: "For the AP $3, 3, 3, \ldots$ the common difference is undefined." The correct response is: 1

(a) $d = 1$ **(b)** Incorrect — $d = 0$

(c) Correct **(d)** $d = 3$

5. The sum of the first 20 odd natural numbers is: 1

(a) 200 **(b)** 420 **(c)** 380 **(d)** 400

6. The distance between $(0, 5)$ and $(-5, 0)$ is: 1

(a) $2\sqrt{5}$ **(b)** $5\sqrt{2}$ **(c)** 10 **(d)** 5

7. The mid-point of the segment joining $(-3, 4)$ and $(5, -2)$ is: 1

(a) $(4, 3)$ **(b)** $(1, 3)$ **(c)** $(-4, 3)$ **(d)** $(1, 1)$

8. The point dividing the join of $(0, 0)$ and $(6, 9)$ in the ratio 1 : 2 is: 1

(a) $(2, 3)$ **(b)** $(4, 6)$ **(c)** $(2, 6)$ **(d)** $(3, 4.5)$

9. A student states: "The points $(1, 1)$, $(2, 2)$ and $(3, 3)$ form a triangle." The correct response is: 1

(a) Incorrect — they are collinear (on $y = x$)

(b) They form a right triangle

(c) Correct

(d) Cannot be determined

10. For the distance of $(3, 4)$ from the x-axis, Student A says 4; Student B says 5. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the nth term of the AP $1, 4, 7, \ldots$ and hence its 25th term. **2**

12. Find the sum of the first 12 terms of the AP $5, 9, 13, \ldots$ **2**

13. If the distance between $(2, 3)$ and $(5, k)$ is 5 units, find k. **2**

14. Find the coordinates of the point dividing the join of A(1, 3) and B(7, 9) in the ratio 1 : 2. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The 8th term of an AP is 31 and its 15th term is 16 more than its 11th term. Find the AP. **3**

16. Find the ratio in which P(−1, 3) divides the segment joining A(−3, 1) and B(5, 9). **3**

17. How many three-digit numbers are divisible by 7? **3**

OR

Find the sum of all multiples of 9 lying between 300 and 700. **3**

18. Show that A(5, 6), B(1, 5), C(2, 1) and D(6, 2) are the vertices of a square. **3**

OR

Find the point on the x-axis equidistant from $(-2, 5)$ and $(2, -3)$. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Ravi saves ₹50 in the first week and increases his saving by ₹20 every week. **5**

(i) How much does he save in the 10th week?

(ii) Find his total saving in 10 weeks.

(iii) In how many weeks will his total saving be ₹2850?

OR

The sum of the first n terms of an AP is $5n^2 + 3n$. Find the AP, its 20th term, and determine which term is 108. **5**

20. **Case study:** Four trees in a garden are at P(−1, 2), Q(3, 2), R(3, −1) and S(−1, −1). **5**

(i) Find PQ and QR.

(ii) Find the diagonals PR and QS.

(iii) What type of quadrilateral is PQRS?

(iv) Find the centre of the plot.

OR

If P$(9a - 2, -b)$ divides the join of A$(3a + 1, -3)$ and B$(8a, 5)$ in the ratio 3 : 1, find a and b. Then find AB. **5**

END OF QUESTION PAPER — **Total: 40 Marks**